

Hospital Supply Chain

Procurement Analytics

1. What is the organization's total procurement spend?

```
SELECT  
    SUM(Total_PO_Value) AS Total_Procurement_Spend  
FROM Purchase_Orders;
```

	Total_Procurement_Spend
1	5201797270.29

2. Which hospitals spend the most?

```
SELECT  
    Hospital_Name,  
    SUM(Total_PO_Value) AS Total_Spend  
FROM vw_Purchase_Order_Summary  
GROUP BY Hospital_Name  
ORDER BY Total_Spend DESC;
```

	Hospital_Name	Total_Spend
1	MedCare Pune	1150406390.84
2	MedCare Mumbai	1108849270.92
3	MedCare Chennai	1078349039.72
4	MedCare Hyderabad	943907139.89
5	MedCare Bangalore	920285428.92

3. Procurement Spend by Department

```
SELECT  
    Department_Name,  
    SUM(Total_PO_Value) AS Total_Spend  
FROM vw_Purchase_Order_Summary  
GROUP BY Department_Name  
ORDER BY Total_Spend DESC;
```

	Department_Name	Total_Spend
1	Procurement	404262245.94
2	Cardiology	397618880.79
3	Gynecology	349549549.33
4	Radiology	330604650.31
5	Administration	329248290.96
6	Emergency	320656903.52
7	Operating Theatre	314295065.62
8	Nephrology	294173304.84
9	Maintenance	286781542.04
10	Dermatology	278166979.97
11	ICU	269415815.29
12	Neurology	251186591.84
13	Pharmacy	231284335.47
14	General Ward	222449675.66
15	ENT	202339475.25
16	Pediatrics	196134457.83
17	Pulmonology	160593261.81
18	Oncology	159480541.94
19	Orthopedics	111567288.97
20	Laboratory	91988412.91

4. Monthly Procurement Trend

```

SELECT
    YEAR(PO_Date) AS Year,
    MONTH(PO_Date) AS Month,
    SUM(Total_PO_Value) AS Monthly_Spend
FROM Purchase_Orders
GROUP BY
    YEAR(PO_Date),
    MONTH(PO_Date)
ORDER BY
    Year,
    Month;

```

	Year	Month	Monthly_Spend
1	2023	1	184039241.71
2	2023	2	84643637.66
3	2023	3	99576055.44
4	2023	4	75675780.26
5	2023	5	186353010.49
6	2023	6	179535308.28
7	2023	7	237840700.93
8	2023	8	267992424.21
9	2023	9	223194455.22
10	2023	10	236508659.60
11	2023	11	47450739.65
12	2023	12	193404780.39
13	2024	1	158251520.62
14	2024	2	99316087.43
15	2024	3	95275350.75
16	2024	4	76869349.84
17	2024	5	90894181.51
18	2024	6	72553903.52
19	2024	7	112951526.66
20	2024	8	150446096.10
21	2024	9	162737778.92
22	2024	10	273274859.15
23	2024	11	166391440.83
24	2024	12	109740506.36
25	2025	1	99777854.42
26	2025	2	209940046.77
27	2025	3	130817985.98
28	2025	4	206327978.83
29	2025	5	186961641.13
30	2025	6	173656416.23
31	2025	7	61985490.73
32	2025	8	66309154.69
33	2025	9	145692108.42
34	2025	10	122689792.70
35	2025	11	56586996.01
36	2025	12	156134408.85

5. Purchase Order Status Distribution

```

SELECT
    PO_Status,
    COUNT(*) AS Total_Orders
FROM Purchase_Orders
GROUP BY PO_Status;

```

	PO_Status	Total_Orders
1	Pending	186
2	In Transit	166
3	Approved	180
4	Delivered	468

6. Average Purchase Order Value

```
SELECT
    AVG(Total_PO_Value) AS Average_PO_Value
FROM Purchase_Orders;
```

	Average_PO_Value
1	5201797.270290

7. Top 10 Highest Value Purchase Orders

```
SELECT TOP (10)
    PO_ID,
    Hospital_Name,
    Vendor_Name,
    Total_PO_Value
FROM vw_Purchase_Order_Summary
ORDER BY Total_PO_Value DESC;
```

	PO_ID	Hospital_Name	Vendor_Name	Total_PO_Value
1	PO000098	MedCare Mumbai	GE HealthCare	64235413.41
2	PO000702	MedCare Chennai	Siemens Healthineers	57670950.48
3	PO000460	MedCare Pune	GE HealthCare	50259715.40
4	PO000876	MedCare Bangalore	Philips Healthcare	47769469.32
5	PO000791	MedCare Pune	Philips Healthcare	46149718.90
6	PO000894	MedCare Chennai	GE HealthCare	45299471.76
7	PO000350	MedCare Chennai	Drager	45002348.22
8	PO000462	MedCare Mumbai	Siemens Healthineers	44279408.37
9	PO000934	MedCare Pune	GE HealthCare	43814700.76
10	PO000863	MedCare Mumbai	Siemens Healthineers	43707378.78

8. Purchase Orders by Vendor

```
SELECT
    Vendor_Name,
    COUNT(*) AS Purchase_Order_Count
FROM vw_Purchase_Order_Summary
GROUP BY Vendor_Name
ORDER BY Purchase_Order_Count DESC;
```

	Vendor_Name	Purchase_Order_Count
1	Thermo Fisher	97
2	Johnson & Johnson MedTech	95
3	Siemens Healthineers	94
4	GE HealthCare	88
5	Baxter India	86
6	Stryker	86
7	Becton Dickinson	80
8	3M India	78
9	Abbott	77
10	Honeywell	76
11	Drager	73
12	Philips Healthcare	70

9. Delivered vs Pending Orders

```

SELECT
    PO_Status,
    COUNT(*) AS Orders
FROM Purchase_Orders
GROUP BY PO_Status;

```

	PO_Status	Orders
1	Pending	186
2	In Transit	166
3	Approved	180
4	Delivered	468

10. Average Delivery Delay

```

SELECT
    AVG(Delivery_Variance_Days) AS Avg_Delivery_Delay
FROM Shipments;

```

	Avg_Delivery_Delay
1	1

Inventory Analytics

1. Total Inventory Value

```
SELECT
    SUM(Inventory_Value) AS Total_Inventory_Value
FROM Inventory;
```

	Total_Inventory_Value
1	13396447640.22

2. Inventory Value by Hospital

```
SELECT
    Hospital_Name,
    SUM(Inventory_Value) AS Inventory_Value
FROM vw_Inventory_Status
GROUP BY Hospital_Name
ORDER BY Inventory_Value DESC;
```

	Hospital_Name	Inventory_Value
1	MedCare Mumbai	3337694560.08
2	MedCare Chennai	2988058117.84
3	MedCare Hyderabad	2554620974.23
4	MedCare Pune	2375162380.30
5	MedCare Bangalore	2140911607.77

3. Products Below Reorder Level

```
SELECT
    Hospital_Name,
    Product_Name,
    Current_Stock,
    Reorder_Level
FROM vw_Inventory_Status
WHERE Current_Stock < Reorder_Level
ORDER BY Hospital_Name;
```

	Hospital_Name	Product_Name	Current_Stock	Reorder_Level
4	MedCare Bangalore	Syringe 5ml	27	37
5	MedCare Bangalore	MRI Scanner	11	35
6	MedCare Bangalore	Patient Monitor	20	107
7	MedCare Bangalore	Patient Monitor	15	58
8	MedCare Bangalore	ECG Machine	13	61
9	MedCare Bangalore	Ventilator	6	155
10	MedCare Bangalore	COVID Test Kit	32	69
11	MedCare Bangalore	MRI Scanner	6	134
12	MedCare Bangalore	IV Cannula	108	179
13	MedCare Bangalore	Blood Collection Tube	114	137
14	MedCare Bangalore	Ventilator	3	132
15	MedCare Bangalore	Ventilator	20	27
16	MedCare Bangalore	MRI Scanner	5	58
17	MedCare Bangalore	Patient Monitor	19	191
18	MedCare Bangalore	Ventilator	17	127
19	MedCare Bangalore	Patient Monitor	18	151
20	MedCare Bangalore	Ventilator	1	89
21	MedCare Bangalore	Syringe 5ml	80	82
22	MedCare Bangalore	Patient Monitor	15	146
23	MedCare Bangalore	MRI Scanner	13	194
24	MedCare Bangalore	Paracetamol	70	116
25	MedCare Bangalore	Ventilator	20	145
26	MedCare Bangalore	ECG Machine	4	149
27	MedCare Bangalore	N95 Mask	95	147
28	MedCare Bangalore	Patient Monitor	3	161
29	MedCare Bangalore	ECG Machine	0	160
30	MedCare Bangalore	Ventilator	13	105
31	MedCare Bangalore	MRI Scanner	20	184
32	MedCare Bangalore	Patient Monitor	17	69
33	MedCare Bangalore	Ventilator	1	75
34	MedCare Bangalore	ECG Machine	9	190
35	MedCare Bangalore	MRI Scanner	19	120
36	MedCare Bangalore	Syringe 5ml	142	163
37	MedCare Bangalore	Patient Monitor	7	99
38	MedCare Bangalore	Ventilator	7	69
39	MedCare Bangalore	Patient Monitor	10	180

4. Critical Products Running Low

```

SELECT
    Hospital_Name,
    Product_Name,
    Current_Stock,
    Safety_Stock
FROM vw_Inventory_Status
WHERE Critical_Item = 'Yes'
AND Current_Stock < Safety_Stock;

```

	Hospital_Name	Product_Name	Current_Stock	Safety_Stock
1	MedCare Bangalore	Ventilator	0	75
2	MedCare Bangalore	Patient Monitor	3	80
3	MedCare Bangalore	Patient Monitor	16	61
4	MedCare Bangalore	Ventilator	6	87
5	MedCare Bangalore	Ventilator	7	51
6	MedCare Bangalore	Patient Monitor	18	61
7	MedCare Bangalore	Ventilator	5	100
8	MedCare Bangalore	Ventilator	10	25
9	MedCare Bangalore	Ventilator	3	33
10	MedCare Bangalore	Patient Monitor	17	41
11	MedCare Bangalore	Patient Monitor	19	48
12	MedCare Bangalore	Patient Monitor	18	83
13	MedCare Bangalore	IV Drip Set	71	87
14	MedCare Bangalore	Ventilator	7	18
15	MedCare Bangalore	Patient Monitor	20	29

5. Inventory by Category

```

SELECT
    Category,
    SUM(Current_Stock) AS Total_Stock,
    SUM(Inventory_Value) AS Total_Value
FROM vw_Inventory_Status
GROUP BY Category
ORDER BY Total_Value DESC;

```

	Category	Total_Stock	Total_Value
1	Equipment	3823	13268121642.09
2	PPE	129898	100772557.58
3	Diagnostics	33156	11207317.27
4	Medicines	79841	8341173.44
5	Consumables	143642	7387069.14
6	Laboratory	41808	617880.70

6. Top 10 Most Valuable Products

```

SELECT TOP (10)
    Product_Name,
    SUM(Inventory_Value) AS Inventory_Value
FROM vw_Inventory_Status
GROUP BY Product_Name
ORDER BY Inventory_Value DESC;

```

	Product_Name	Inventory_Value
1	MRI Scanner	11882795318.20
2	Ventilator	982637324.21
3	Patient Monitor	230350065.51
4	ECG Machine	172338934.17
5	N95 Mask	57593896.68
6	Surgical Mask	22785849.87
7	Latex Gloves	20392811.03
8	COVID Test Kit	11207317.27
9	Amoxicillin	6620051.62
10	IV Drip Set	4674710.24

7. Stock Status Distribution

```
SELECT
    Stock_Status,
    COUNT(*) AS Products
FROM Inventory
GROUP BY Stock_Status;
```

	Stock_Status	Products
1	In Stock	787
2	Low Stock	443
3	Out of Stock	20

8. Hospitals with Lowest Stock Levels

```
SELECT
    Hospital_Name,
    SUM(Current_Stock) AS Total_Stock
FROM vw_Inventory_Status
GROUP BY Hospital_Name
ORDER BY Total_Stock;
```

	Hospital_Name	Total_Stock
1	MedCare Chennai	79905
2	MedCare Pune	80424
3	MedCare Mumbai	86153
4	MedCare Bangalore	89969
5	MedCare Hyderabad	95717

9. Disposable vs Non-Disposable Inventory

```
SELECT  
    Disposable,  
    SUM(Current_Stock) AS Total_Stock,  
    SUM(Inventory_Value) AS Total_Value  
FROM vw_Inventory_Status  
GROUP BY Disposable;
```

	Disposable	Total_Stock	Total_Value
1	Yes	428345	128325998.13
2	No	3823	13268121642.09

10. Highest Stocked Products

```
SELECT TOP (10)  
    Product_Name,  
    SUM(Current_Stock) AS Total_Stock  
FROM vw_Inventory_Status  
GROUP BY Product_Name  
ORDER BY Total_Stock DESC;
```

	Product_Name	Total_Stock
1	IV Drip Set	56684
2	N95 Mask	45024
3	IV Cannula	44285
4	Paracetamol	43218
5	Latex Gloves	42688
6	Syringe 5ml	42673
7	Surgical Mask	42186
8	Blood Collection Tube	41808
9	Amoxicillin	36623
10	COVID Test Kit	33156

Vendor Analytics

1. Which vendors account for the highest procurement spend?

```
SELECT
    Vendor_Name,
    SUM(Total_PO_Value) AS Total_Spend,
    RANK() OVER (
        ORDER BY SUM(Total_PO_Value) DESC
    ) AS Vendor_Rank
FROM vw_Purchase_Order_Summary
GROUP BY Vendor_Name;
```

	Vendor_Name	Total_Spend	Vendor_Rank
1	Siemens Healthineers	1508560983.37	1
2	GE HealthCare	1446809201.58	2
3	Drager	1155998901.79	3
4	Philips Healthcare	1051773530.64	4
5	3M India	13084270.62	5
6	Honeywell	13074954.55	6
7	Abbott	6019239.36	7
8	Johnson & Johnson MedTech	1929569.97	8
9	Stryker	1548685.08	9
10	Becton Dickinson	1310072.63	10
11	Baxter India	975297.75	11
12	Thermo Fisher	712562.95	12

2. Top 5 Vendors

```
WITH VendorSpend AS
(
    SELECT
        Vendor_Name,
        SUM(Total_PO_Value) AS Total_Spend,
        RANK() OVER(
            ORDER BY SUM(Total_PO_Value) DESC
        ) AS Vendor_Rank
    FROM vw_Purchase_Order_Summary
    GROUP BY Vendor_Name
)
SELECT *
FROM VendorSpend
WHERE Vendor_Rank <= 5;
```

	Vendor_Name	Total_Spend	Vendor_Rank
1	Siemens Healthineers	1508560983.37	1
2	GE HealthCare	1446809201.58	2
3	Drager	1155998901.79	3
4	Philips Healthcare	1051773530.64	4
5	3M India	13084270.62	5

3. Vendor Delivery Performance

```
SELECT
    Vendor_Name,
    AVG(Avg_Delivery_Variance) AS Avg_Delay
FROM vw_Vendor_Performance
GROUP BY Vendor_Name
ORDER BY Avg_Delay;
```

	Vendor_Name	Avg_Delay
1	3M India	1
2	Abbott	1
3	Baxter India	1
4	Becton Dickinson	1
5	Drager	1
6	GE HealthCare	1
7	Honeywell	1
8	Johnson & Johnson MedTech	1
9	Philips Healthcare	1
10	Siemens Healthineers	1
11	Stryker	1
12	Thermo Fisher	1

4. Vendors with Late Deliveries

```
SELECT
    Vendor_Name,
    Avg_Delivery_Variance
FROM vw_Vendor_Performance
WHERE Avg_Delivery_Variance > 2
ORDER BY Avg_Delivery_Variance DESC;
```

Vendor_Name	Avg_Delivery_Variance

5. Vendor Product Coverage

```
SELECT
    Vendor_Name,
    Products_Supplied
FROM vw_Vendor_Performance
ORDER BY Products_Supplied DESC;
```

	Vendor_Name	Products_Supplied
1	Becton Dickinson	113
2	Baxter India	108
3	3M India	97
4	Honeywell	97
5	Drager	88
6	Siemens Healthineers	86
7	Philips Healthcare	79
8	GE HealthCare	78
9	Abbott	40
10	Thermo Fisher	38
11	Johnson & Johnson MedTech	12
12	Stryker	10

6. Vendor Rating vs Spend

```

SELECT
    Vendor_Name,
    Vendor_Rating,
    Total_Spend
FROM vw_Vendor_Performance
ORDER BY Vendor_Rating DESC,
         Total_Spend DESC;

```

	Vendor_Name	Vendor_Rating	Total_Spend
1	Siemens Healthineers	4.9	129736244569.82
2	Philips Healthcare	4.9	83090108920.56
3	Becton Dickinson	4.9	148038207.19
4	Johnson & Johnson MedTech	4.9	23154839.64
5	GE HealthCare	4.8	112851117723.24
6	Drager	4.8	101727903357.52
7	3M India	4.8	1269174250.14
8	Baxter India	4.8	105332157.00
9	Thermo Fisher	4.8	27077392.10
10	Stryker	4.8	15486850.80
11	Honeywell	4.7	1268270591.35
12	Abbott	4.7	240769574.40

7. Vendor Lead Time Analysis

```

SELECT
    Vendor_Name,
    Lead_Time_Days,
    DENSE_RANK() OVER(
        ORDER BY Lead_Time_Days
    ) AS Lead_Time_Rank
FROM vw_Vendor_Performance;

```

	Vendor_Name	Lead_Time_Days	Lead_Time_Rank
1	Becton Dickinson	5	1
2	Baxter India	6	2
3	3M India	7	3
4	Honeywell	10	4
5	Abbott	12	5
6	Thermo Fisher	14	6
7	Johnson & Johnson MedTech	15	7
8	Stryker	16	8
9	GE HealthCare	18	9
10	Philips Healthcare	21	10
11	Drager	22	11
12	Siemens Healthineers	25	12

8. Preferred Vendors

```
SELECT *
FROM vw_Vendor_Performance
WHERE Preferred_Vendor='Yes'
ORDER BY Vendor_Rating DESC;
```

Vendor_ID	Vendor_Name	Vendor_Type	Primary_Category	Vendor_Rating	Quality_Score	On_Time_Delivery_Percentage	Lead_Time_Days	Preferred_Vendor	Products_Supplied	Total_Purchase_Orders	Total_Spend	Avg_Delivery_Variance	Total_Shipments
1	V005	Philips Healthcare	Manufacturer	4.9	98.90	96.40	21	Yes	79	70	83090108920.56	1	70
2	V007	Siemens Healthineers	Manufacturer	4.9	98.00	96.90	25	Yes	86	94	129736244569.82	1	94
3	V011	Johnson & Johnson MedTech	Manufacturer	4.9	98.40	99.00	15	Yes	12	95	23154839.64	1	95
4	V003	Becton Dickinson	Manufacturer	4.9	94.50	99.10	5	Yes	113	80	148038207.19	1	80
5	V010	Thermo Fisher	Manufacturer	4.8	94.10	98.20	14	Yes	38	97	27077392.10	1	97
6	V004	Baxter India	Manufacturer	4.8	95.60	98.00	6	Yes	108	86	105332157.00	1	86
7	V001	3M India	Manufacturer	4.8	99.90	98.50	7	Yes	97	78	1269174250.14	1	78
8	V012	Stryker	Manufacturer	4.8	95.90	98.60	16	Yes	10	86	15486850.80	1	86
9	V006	GE HealthCare	Manufacturer	4.8	94.30	97.80	18	Yes	78	88	112851117723.24	1	88
10	V008	Drager	Manufacturer	4.8	95.60	97.10	22	Yes	88	73	101727903357.52	1	73
11	V002	Honeywell	Manufacturer	4.7	99.10	97.20	10	Yes	97	76	1268270591.35	1	76
12	V009	Abbott	Manufacturer	4.7	99.40	98.40	12	Yes	40	77	240769574.40	1	77

9. Vendor Contribution %

```
SELECT
    Vendor_Name,
    Total_Spend,
    ROUND(
        Total_Spend * 100.0 /
        SUM(Total_Spend) OVER(),
        2
    ) AS Spend_Percentage
FROM vw_Vendor_Performance
ORDER BY Spend_Percentage DESC;
```

	Vendor_Name	Total_Spend	Spend_Percentage
1	Siemens Healthineers	129736244569.82	30.140000
2	GE HealthCare	112851117723.24	26.210000
3	Drager	101727903357.52	23.630000
4	Philips Healthcare	83090108920.56	19.300000
5	3M India	1269174250.14	0.290000
6	Honeywell	1268270591.35	0.290000
7	Abbott	240769574.40	0.060000
8	Becton Dickinson	148038207.19	0.030000
9	Baxter India	105332157.00	0.020000
10	Johnson & Johnson MedTech	23154839.64	0.010000
11	Thermo Fisher	27077392.10	0.010000
12	Stryker	15486850.80	0.000000

10. Vendor Scorecard

SELECT

```

Vendor_Name,
Vendor_Rating,
Quality_Score,
On_Time_Delivery_Percentage,
Lead_Time_Days,
Total_Spend,
Vendor_Rank =
RANK() OVER(
    ORDER BY Vendor_Rating DESC,
             Total_Spend DESC
)

```

FROM vw_Vendor_Performance;

	Vendor_Name	Vendor_Rating	Quality_Score	On_Time_Delivery_Percentage	Lead_Time_Days	Total_Spend	Vendor_Rank
1	Siemens Healthineers	4.9	98.00	96.90	25	129736244569.82	1
2	Philips Healthcare	4.9	98.90	96.40	21	83090108920.56	2
3	Becton Dickinson	4.9	94.50	99.10	5	148038207.19	3
4	Johnson & Johnson MedTech	4.9	98.40	99.00	15	23154839.64	4
5	GE HealthCare	4.8	94.30	97.80	18	112851117723.24	5
6	Drager	4.8	95.60	97.10	22	101727903357.52	6
7	3M India	4.8	99.50	98.50	7	1269174250.14	7
8	Baxter India	4.8	95.60	98.00	6	105332157.00	8
9	Thermo Fisher	4.8	94.10	98.20	14	27077392.10	9
10	Stryker	4.8	95.90	98.60	16	15486850.80	10
11	Honeywell	4.7	99.10	97.20	10	1268270591.35	11
12	Abbott	4.7	99.40	98.40	12	240769574.40	12

Financial Analytics

1. Total Organizational Spend

```
SELECT  
    SUM(Actual_Amount) AS Total_Spend  
FROM Financials;
```

	Total_Spend
1	5201797270.29

2. Budget vs Actual by Hospital

```
SELECT  
    Hospital_Name,  
    SUM(Budget_Amount) AS Budget,  
    SUM(Actual_Amount) AS Actual,  
    SUM(Variance) AS Variance  
FROM vw_Financial_Summary  
GROUP BY Hospital_Name  
ORDER BY Actual DESC;
```

	Hospital_Name	Budget	Actual	Variance
1	MedCare Pune	1164675088.99	1150406390.84	-14268698.15
2	MedCare Mumbai	1091544476.99	1108849270.92	17304793.93
3	MedCare Chennai	1087781518.88	1078349039.72	-9432479.16
4	MedCare Hyderabad	951445075.35	943907139.89	-7537935.46
5	MedCare Bangalore	920343035.86	920285428.92	-57606.94

3. Budget Utilization %

```
SELECT  
    Hospital_Name,  
    SUM(Actual_Amount) AS Actual,  
    SUM(Budget_Amount) AS Budget,  
    ROUND(  
        SUM(Actual_Amount) * 100.0 /  
        SUM(Budget_Amount),  
        2  
    ) AS Budget_Utilization_Percentage  
FROM vw_Financial_Summary  
GROUP BY Hospital_Name;
```

	Hospital_Name	Actual	Budget	Budget_Utilization_Percentage
1	MedCare Bangalore	920285428.92	920343035.86	99.990000
2	MedCare Chennai	1078349039.72	1087781518.88	99.130000
3	MedCare Hyderabad	943907139.89	951445075.35	99.210000
4	MedCare Mumbai	1108849270.92	1091544476.99	101.590000
5	MedCare Pune	1150406390.84	1164675088.99	98.770000

4. Department Spend Ranking

```

SELECT
    Department_Name,
    SUM(Actual_Amount) AS Total_Spend,
    RANK() OVER(
        ORDER BY SUM(Actual_Amount) DESC
    ) AS Spend_Rank
FROM vw_Financial_Summary
GROUP BY Department_Name;

```

	Department_Name	Total_Spend	Spend_Rank
1	Procurement	404262245.94	1
2	Cardiology	397618880.79	2
3	Gynecology	349549549.33	3
4	Radiology	330604650.31	4
5	Administration	329248290.96	5
6	Emergency	320656903.52	6
7	Operating Theatre	314295065.62	7
8	Nephrology	294173304.84	8
9	Maintenance	286781542.04	9
10	Dermatology	278166979.97	10
11	ICU	269415815.29	11
12	Neurology	251186591.84	12
13	Pharmacy	231284335.47	13
14	General Ward	222449675.66	14
15	ENT	202339475.25	15
16	Pediatrics	196134457.83	16
17	Pulmonology	160593261.81	17
18	Oncology	159480541.94	18
19	Orthopedics	111567288.97	19
20	Laboratory	91988412.91	20

5. Quarterly Spending Trend

```
SELECT
    Fiscal_Year,
    Fiscal_Quarter,
    SUM(Actual_Amount) AS Quarterly_Spend
FROM vw_Financial_Summary
GROUP BY
    Fiscal_Year,
    Fiscal_Quarter
ORDER BY
    Fiscal_Year,
    Fiscal_Quarter;
```

	Fiscal_Year	Fiscal_Quarter	Quarterly_Spend
1	FY2023	Q1	334103814.81
2	FY2023	Q2	312369890.98
3	FY2023	Q3	807480133.29
4	FY2023	Q4	392865239.06
5	FY2024	Q1	457732366.26
6	FY2024	Q2	232723944.57
7	FY2024	Q3	354017645.47
8	FY2024	Q4	584955115.59
9	FY2025	Q1	462882847.54
10	FY2025	Q2	512798959.79
11	FY2025	Q3	315301669.09
12	FY2025	Q4	332563645.05
13	FY2026	Q1	102001998.79

6. Pending Payments

```
SELECT
    Vendor_Name,
    Invoice_Number,
    Actual_Amount,
    Payment_Due_Date,
    DATEDIFF(
        DAY,
        Payment_Due_Date,
        GETDATE()
    ) AS Days_Overdue
FROM vw_Financial_Summary
WHERE Payment_Status = 'Pending'
ORDER BY Payment_Due_Date;
```

	Vendor_Name	Invoice_Number	Actual_Amount	Payment_Due_Date	Days_Overdue
1	GE HealthCare	INV-2023-000683	13737767.79	2023-03-27	1243
2	Johnson & Johnson MedTech	INV-2023-000276	25091.45	2023-03-30	1240
3	3M India	INV-2023-000361	183993.36	2023-04-08	1231
4	Philips Healthcare	INV-2023-000585	24877969.47	2023-04-29	1210
5	Drager	INV-2023-000908	15079709.69	2023-05-15	1194
6	Honeywell	INV-2023-000703	312832.73	2023-05-19	1190
7	Baxter India	INV-2023-000205	17690.50	2023-06-12	1166
8	Baxter India	INV-2023-000948	10537.84	2023-06-26	1152
9	Stryker	INV-2023-000824	5716.93	2023-07-13	1135
10	Thermo Fisher	INV-2023-000652	5628.55	2023-07-16	1132
11	Thermo Fisher	INV-2023-000729	7604.52	2023-07-19	1129
12	Thermo Fisher	INV-2023-000393	1842.22	2023-07-20	1128
13	Philips Healthcare	INV-2023-000188	30392532.75	2023-07-22	1126
14	Becton Dickinson	INV-2023-000371	15593.28	2023-07-25	1123
15	Thermo Fisher	INV-2023-000843	11982.76	2023-07-27	1121
16	Johnson & Johnson MedTech	INV-2023-000024	33501.78	2023-07-30	1118
17	Siemens Healthineers	INV-2023-000677	17158143.64	2023-07-30	1118
18	Siemens Healthineers	INV-2023-000151	24713742.29	2023-08-04	1113
19	Becton Dickinson	INV-2023-000427	14304.61	2023-08-12	1105
20	GE HealthCare	INV-2023-000934	43814700.76	2023-08-17	1100
21	Johnson & Johnson MedTech	INV-2023-000982	35600.28	2023-09-01	1085
22	3M India	INV-2023-000441	477417.25	2023-09-03	1083
23	Becton Dickinson	INV-2023-000001	15605.66	2023-09-19	1067
24	Baxter India	INV-2023-000666	17340.59	2023-09-23	1063
25	Honeywell	INV-2023-000513	299585.36	2023-10-02	1054
26	Stryker	INV-2023-000162	23307.63	2023-10-06	1050
27	Abbott	INV-2023-000830	84711.12	2023-10-10	1046
28	Baxter India	INV-2023-000865	16025.17	2023-10-13	1043
29	Johnson & Johnson MedTech	INV-2023-000796	28787.32	2023-10-30	1026
30	GE HealthCare	INV-2023-000626	7480384.02	2023-11-15	1010
31	Drager	INV-2023-000548	35514923.89	2023-11-20	1005
32	Baxter India	INV-2023-000727	7650.98	2023-11-21	1004
33	Johnson & Johnson MedTech	INV-2023-000469	32518.91	2023-11-23	1002
34	GE HealthCare	INV-2023-000894	45299471.76	2023-11-24	1001
35	Philips Healthcare	INV-2023-000271	24737764.78	2023-12-04	991
36	Becton Dickinson	INV-2023-000926	22084.66	2023-12-05	990
37	Johnson & Johnson MedTech	INV-2023-000433	32042.33	2023-12-06	989
38	Johnson & Johnson MedTech	INV-2023-000844	87171.15	2023-12-18	978

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7. Top 10 Largest Invoices

```

SELECT TOP (10)
    Invoice_Number,
    Vendor_Name,
    Hospital_Name,
    Actual_Amount,
    RANK() OVER(
        ORDER BY Actual_Amount DESC
    ) AS Invoice_Rank
FROM vw_Financial_Summary;

```

	Invoice_Number	Vendor_Name	Hospital_Name	Actual_Amount	Invoice_Rank
1	INV-2024-000098	GE HealthCare	MedCare Mumbai	64235413.41	1
2	INV-2024-000702	Siemens Healthineers	MedCare Chennai	57670950.48	2
3	INV-2024-000460	GE HealthCare	MedCare Pune	50259715.40	3
4	INV-2025-000876	Philips Healthcare	MedCare Bangalore	47769469.32	4
5	INV-2023-000791	Philips Healthcare	MedCare Pune	46149718.90	5
6	INV-2023-000894	GE HealthCare	MedCare Chennai	45299471.76	6
7	INV-2023-000350	Drager	MedCare Chennai	45002348.22	7
8	INV-2023-000462	Siemens Healthineers	MedCare Mumbai	44279408.37	8
9	INV-2023-000934	GE HealthCare	MedCare Pune	43814700.76	9
10	INV-2023-000863	Siemens Healthineers	MedCare Mumbai	43707378.78	10

8. Expense Category Analysis

```

SELECT
    Expense_Category,
    COUNT(*) AS Transactions,
    SUM(Actual_Amount) AS Total_Spend,
    AVG(Actual_Amount) AS Avg_Spend
FROM vw_Financial_Summary
GROUP BY Expense_Category
ORDER BY Total_Spend DESC;

```

	Expense_Category	Transactions	Total_Spend	Avg_Spend
1	Equipment	201	1178975489.67	5865549.699850
2	Laboratory	201	1125752741.16	5600759.906268
3	Administration	217	1042498535.81	4804140.718018
4	Pharmacy	199	946927654.10	4758430.422613
5	Medical Supplies	182	907642849.55	4987048.623901

9. Vendor Payment Analysis

```

SELECT
    Vendor_Name,
    SUM(Actual_Amount) AS Total_Paid,
    COUNT(*) AS Invoice_Count,
    DENSE_RANK() OVER(
        ORDER BY SUM(Actual_Amount) DESC
    ) AS Vendor_Rank
FROM vw_Financial_Summary
GROUP BY Vendor_Name;

```

	Vendor_Name	Total_Paid	Invoice_Count	Vendor_Rank
1	Siemens Healthineers	1508560983.37	94	1
2	GE HealthCare	1446809201.58	88	2
3	Drager	1155998901.79	73	3
4	Philips Healthcare	1051773530.64	70	4
5	3M India	13084270.62	78	5
6	Honeywell	13074954.55	76	6
7	Abbott	6019239.36	77	7
8	Johnson & Johnson MedTech	1929569.97	95	8
9	Stryker	1548685.08	86	9
10	Becton Dickinson	1310072.63	80	10
11	Baxter India	975297.75	86	11
12	Thermo Fisher	712562.95	97	12

10. Running Spend by Invoice Date

```

SELECT
    Invoice_Date,
    SUM(Actual_Amount) AS Daily_Spend,
    SUM(SUM(Actual_Amount))
    OVER(
        ORDER BY Invoice_Date
    ) AS Running_Total

FROM vw_Financial_Summary
GROUP BY Invoice_Date
ORDER BY Invoice_Date;

```

	Invoice_Date	Daily_Spend	Running_Total
1	2023-01-12	22841.87	22841.87
2	2023-01-18	350812.31	373654.18
3	2023-01-20	14293.24	387947.42
4	2023-01-23	53945.79	441893.21
5	2023-01-25	264476.62	706369.83
6	2023-01-26	13737767.79	14444137.62
7	2023-01-27	11699.20	14455836.82
8	2023-01-28	49657212.94	64113049.76
9	2023-01-29	25091.45	64138141.21
10	2023-01-30	307591.16	64445732.37
11	2023-02-01	37285782.77	101731515.14
12	2023-02-03	24677806.77	126409321.91
13	2023-02-04	32600.07	126441921.98
14	2023-02-05	9163933.49	135605855.47
15	2023-02-06	3645622.71	139251478.18
16	2023-02-07	2870.40	139254348.58
17	2023-02-08	19975.01	139274323.59
18	2023-02-10	69083.53	139343407.12
19	2023-02-11	9943847.02	149287254.14
20	2023-02-15	77092.69	149364346.83
21	2023-02-16	1643.75	149365990.58
22	2023-02-17	24239899.15	173605889.73
23	2023-02-21	10524140.00	184130029.73
24	2023-02-28	24983253.72	209113283.45
25	2023-03-01	136061.35	209249344.80
26	2023-03-03	19464.11	209268808.91
27	2023-03-05	103078.20	209371887.11
28	2023-03-06	260616.26	209632503.37
29	2023-03-07	41291291.72	250923795.09
30	2023-03-09	213756.19	251137551.28
31	2023-03-10	75245.23	251212796.51
32	2023-03-12	8251.11	251221047.62
33	2023-03-15	3603.07	251224650.69
34	2023-03-16	15790589.98	267015240.67
35	2023-03-17	17573.87	267032814.54
36	2023-03-19	32872.92	267065687.46
37	2023-03-21	13894.21	267079581.67
38	2023-03-22	9335.22	267088916.89

Hospital Operations

1. Total Patients by Hospital

```
SELECT  
    Hospital_Name,  
    Total_Patients  
FROM vw_Hospital_Operations  
ORDER BY Total_Patients DESC;
```

	Hospital_Name	Total_Patients
1	MedCare Mumbai	1028
2	MedCare Hyderabad	1015
3	MedCare Pune	1000
4	MedCare Bangalore	981
5	MedCare Chennai	976

2. Average Length of Stay Ranking

```
SELECT  
    Hospital_Name,  
    Avg_Length_of_Stay,  
    RANK() OVER(  
        ORDER BY Avg_Length_of_Stay DESC  
    ) AS Stay_Rank  
FROM vw_Hospital_Operations;
```

	Hospital_Name	Avg_Length_of_Stay	Stay_Rank
1	MedCare Chennai	8.13517345684847	1
2	MedCare Bangalore	7.96782200974533	2
3	MedCare Pune	7.9340884338924	3
4	MedCare Hyderabad	7.85593098490295	4
5	MedCare Mumbai	7.83642296415398	5

3. Staff Distribution

```
SELECT  
    Hospital_Name,  
    Total_Staff  
FROM vw_Hospital_Operations  
ORDER BY Total_Staff DESC;
```

	Hospital_Name	Total_Staff
1	MedCare Pune	451
2	MedCare Bangalore	441
3	MedCare Chennai	437
4	MedCare Mumbai	428
5	MedCare Hyderabad	404

4. Staff Experience Ranking

```
SELECT
    Hospital_Name,
    Avg_Staff_Experience,
    DENSE_RANK() OVER(
        ORDER BY Avg_Staff_Experience DESC
    ) AS Experience_Rank
FROM vw_Hospital_Operations;
```

	Hospital_Name	Avg_Staff_Experience	Experience_Rank
1	MedCare Bangalore	11.1846097438013	1
2	MedCare Mumbai	10.9578267835265	2
3	MedCare Pune	10.5823664885065	3
4	MedCare Hyderabad	10.4673549373865	4
5	MedCare Chennai	10.4187591275263	5

5. Staff Performance Ranking

```
SELECT
    Hospital_Name,
    Avg_Staff_Performance,
    RANK() OVER(
        ORDER BY Avg_Staff_Performance DESC
    ) AS Performance_Rank
FROM vw_Hospital_Operations;
```

	Hospital_Name	Avg_Staff_Performance	Performance_Rank
1	MedCare Bangalore	4.03017095192272	1
2	MedCare Mumbai	4.00884459523126	2
3	MedCare Chennai	3.9955528336552	3
4	MedCare Hyderabad	3.97362106873133	4
5	MedCare Pune	3.96297744752971	5

6. Department-wise Patient Count

```
SELECT
    d.Department_Name,
    COUNT(p.Patient_ID) AS Patient_Count
FROM Patients p
JOIN Departments d
ON p.Department_ID = d.Department_ID
GROUP BY d.Department_Name
ORDER BY Patient_Count DESC;
```

	Department_Name	Patient_Count
1	Dermatology	319
2	Gynecology	318
3	Pharmacy	315
4	Nephrology	312
5	ICU	311
6	Neurology	303
7	Pediatrics	300
8	General Ward	299
9	Laboratory	298
10	Orthopedics	295
11	Emergency	289
12	Operating Theatre	288
13	Radiology	288
14	Oncology	280
15	ENT	271
16	Pulmonology	258
17	Cardiology	256

7. Gender Distribution

```
SELECT
    Gender,
    COUNT(*) AS Total_Patients,
    ROUND(
        COUNT(*) * 100.0 /
        SUM(COUNT(*)) OVER(),
        2
    ) AS Percentage
FROM Patients
GROUP BY Gender;
```

	Gender	Total_Patients	Percentage
1	Male	2480	49.600000000000
2	Female	2520	50.400000000000

8. Average Age by Department

```
SELECT
    d.Department_Name,
    AVG(p.Age) AS Average_Age
FROM Patients p
JOIN Departments d
ON p.Department_ID = d.Department_ID
GROUP BY d.Department_Name
ORDER BY Average_Age DESC;
```

	Department_Name	Average_Age
1	Emergency	50
2	Dermatology	49
3	ICU	49
4	Orthopedics	49
5	Pulmonology	49
6	ENT	48
7	Pharmacy	48
8	General Ward	47
9	Pediatrics	47
10	Neurology	47
11	Nephrology	46
12	Operating Theatre	46
13	Radiology	46
14	Gynecology	46
15	Cardiology	46
16	Laboratory	45
17	Oncology	45

9. Patient Outcomes

```
SELECT
    Outcome,
    COUNT(*) AS Total_Patients,
    ROUND(
        COUNT(*) * 100.0 /
        SUM(COUNT(*)) OVER(),
        2
    ) AS Percentage
FROM Patients
GROUP BY Outcome
ORDER BY Total_Patients DESC;
```

	Outcome	Total_Patients	Percentage
1	Recovered	1676	33.520000000000
2	Transferred	1663	33.260000000000
3	Deceased	1661	33.220000000000

10. Hospital Performance Scorecard

```

SELECT
    Hospital_Name,
    Total_Patients,
    Total_Staff,
    Avg_Length_of_Stay,
    Avg_Staff_Experience,
    Avg_Staff_Performance,
    RANK() OVER(
        ORDER BY
            Total_Patients DESC,
            Avg_Staff_Performance DESC
    ) AS Overall_Rank
FROM vw_Hospital_Operations;

```

	Hospital_Name	Total_Patients	Total_Staff	Avg_Length_of_Stay	Avg_Staff_Experience	Avg_Staff_Performance	Overall_Rank
1	MedCare Mumbai	1028	428	7.83642296415398	10.9578267835265	4.00884459523126	1
2	MedCare Hyderabad	1015	404	7.85593098490295	10.4673549373865	3.97362106873133	2
3	MedCare Pune	1000	451	7.9340884338924	10.5823664885065	3.96297744752971	3
4	MedCare Bangalore	981	441	7.96782200974533	11.1846097438013	4.03017095192272	4
5	MedCare Chennai	976	437	8.13517345684847	10.4187591275263	3.9955528336552	5